In [2]:

```python
# Importing the necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Loading the dataset
data = pd.read_csv("airline_customer_satisfaction.csv")

# Quick overview of the dataset
print(data.head())
print(data.info())

# Checking for missing values
missing_data = data.isnull().sum()
print("\nMissing values:\n", missing_data)

# Removing any rows with missing data (if any)
data.dropna(inplace=True)

# Summary statistics
print("\nSummary statistics:\n", data.describe())

# Distribution of satisfaction levels
plt.figure(figsize=(10, 6))
sns.countplot(data=data, x="satisfaction")
plt.title("Distribution of Customer Satisfaction")
plt.show()

# Analyzing factors influencing satisfaction
# List of features
features = [
    "Inflight wifi service", "Departure/Arrival time convenient", "Ease of Online book
    "Food and drink", "Online boarding", "Seat comfort", "Inflight entertainment", "Or
    "Leg room service", "Baggage handling", "Checkin service", "Inflight service", "Cl
    "Departure Delay in Minutes", "Arrival Delay in Minutes"
]

# Correlation heatmap
correlation_matrix = data[features].corr()
plt.figure(figsize=(15, 10))
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix of Features")
plt.show()

# Analyzing each feature against satisfaction using count plots for categorical/ordina
# and boxplots for numerical features
for feature in features:
    plt.figure(figsize=(10, 6))
    if data[feature].dtype == 'O':  # Check if the feature is categorical
        sns.countplot(data=data, x=feature, hue="satisfaction")
    else:  # If the feature is numerical
        sns.boxplot(data=data, x="satisfaction", y=feature)
    plt.title(f"Satisfaction by {feature}")
    plt.show()

# Concluding the EDA
```

```
# From this EDA, you can identify the key factors that influence customer satisfactio
# This will help in pinpointing areas of improvement for the airline.
```

```
   Unnamed: 0     id  Gender      Customer Type  Age   Type of Travel  \
0           0  19556  Female     Loyal Customer   52  Business travel
1           1  90035  Female     Loyal Customer   36  Business travel
2           2  12360    Male  disloyal Customer   20  Business travel
3           3  77959    Male     Loyal Customer   44  Business travel
4           4  36875  Female     Loyal Customer   49  Business travel

      Class  Flight Distance  Inflight wifi service  \
0       Eco              160                      5
1  Business             2863                      1
2       Eco              192                      2
3  Business             3377                      0
4       Eco             1182                      2

   Departure/Arrival time convenient  ...  Inflight entertainment  \
0                                  4  ...                       5
1                                  1  ...                       4
2                                  0  ...                       2
3                                  0  ...                       1
4                                  3  ...                       2

   On-board service  Leg room service  Baggage handling  Checkin service  \
0                 5                 5                 5                2
1                 4                 4                 4                3
2                 4                 1                 3                2
3                 1                 1                 1                3
4                 2                 2                 2                4

   Inflight service  Cleanliness  Departure Delay in Minutes  \
0                 5            5                          50
1                 4            5                           0
2                 2            2                           0
3                 1            4                           0
4                 2            4                           0

   Arrival Delay in Minutes             satisfaction
0                      44.0                satisfied
1                       0.0                satisfied
2                       0.0  neutral or dissatisfied
3                       6.0                satisfied
4                      20.0                satisfied

[5 rows x 25 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25976 entries, 0 to 25975
Data columns (total 25 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   Unnamed: 0                         25976 non-null  int64
 1   id                                 25976 non-null  int64
 2   Gender                             25976 non-null  object
 3   Customer Type                      25976 non-null  object
 4   Age                                25976 non-null  int64
 5   Type of Travel                     25976 non-null  object
 6   Class                              25976 non-null  object
 7   Flight Distance                    25976 non-null  int64
 8   Inflight wifi service              25976 non-null  int64
 9   Departure/Arrival time convenient  25976 non-null  int64
 10  Ease of Online booking             25976 non-null  int64
 11  Gate location                      25976 non-null  int64
```

```
 12  Food and drink                     25976 non-null  int64
 13  Online boarding                    25976 non-null  int64
 14  Seat comfort                       25976 non-null  int64
 15  Inflight entertainment             25976 non-null  int64
 16  On-board service                   25976 non-null  int64
 17  Leg room service                   25976 non-null  int64
 18  Baggage handling                   25976 non-null  int64
 19  Checkin service                    25976 non-null  int64
 20  Inflight service                   25976 non-null  int64
 21  Cleanliness                        25976 non-null  int64
 22  Departure Delay in Minutes         25976 non-null  int64
 23  Arrival Delay in Minutes           25893 non-null  float64
 24  satisfaction                       25976 non-null  object
dtypes: float64(1), int64(19), object(5)
memory usage: 5.0+ MB
None

Missing values:
 Unnamed: 0                            0
id                                    0
Gender                                0
Customer Type                         0
Age                                   0
Type of Travel                        0
Class                                 0
Flight Distance                       0
Inflight wifi service                 0
Departure/Arrival time convenient     0
Ease of Online booking                0
Gate location                         0
Food and drink                        0
Online boarding                       0
Seat comfort                          0
Inflight entertainment                0
On-board service                      0
Leg room service                      0
Baggage handling                      0
Checkin service                       0
Inflight service                      0
Cleanliness                           0
Departure Delay in Minutes            0
Arrival Delay in Minutes             83
satisfaction                          0
dtype: int64

Summary statistics:
          Unnamed: 0             id           Age  Flight Distance  \
count  25893.000000   25893.000000  25893.000000     25893.000000
mean   12987.838566   65021.974858     39.621983      1193.753254
std     7499.175165   37606.098635     15.134224       998.626779
min        0.000000      17.000000      7.000000        31.000000
25%     6496.000000   32209.000000     27.000000       414.000000
50%    12984.000000   65344.000000     40.000000       849.000000
75%    19482.000000   97623.000000     51.000000      1744.000000
max    25975.000000  129877.000000     85.000000      4983.000000

       Inflight wifi service  Departure/Arrival time convenient  \
count           25893.000000                       25893.000000
mean                2.723709                           3.046422
std                 1.334711                           1.532971
```

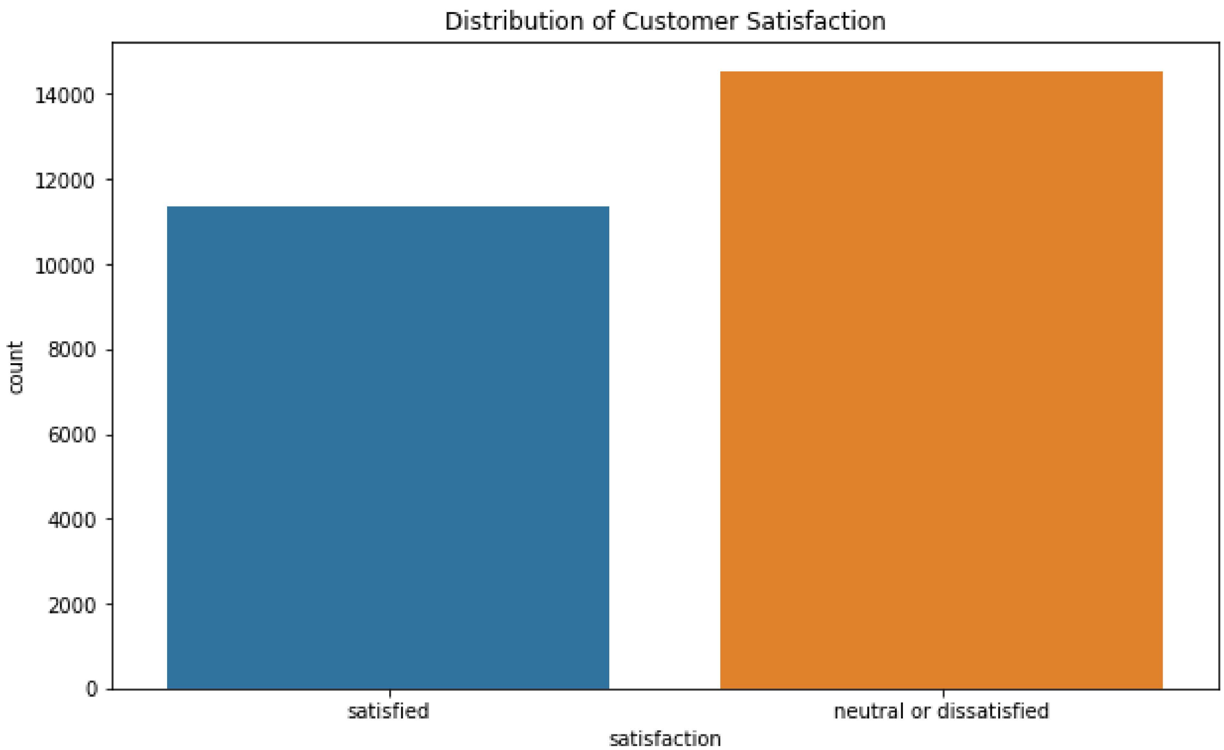
min	0.000000	0.000000
25%	2.000000	2.000000
50%	3.000000	3.000000
75%	4.000000	4.000000
max	5.000000	5.000000

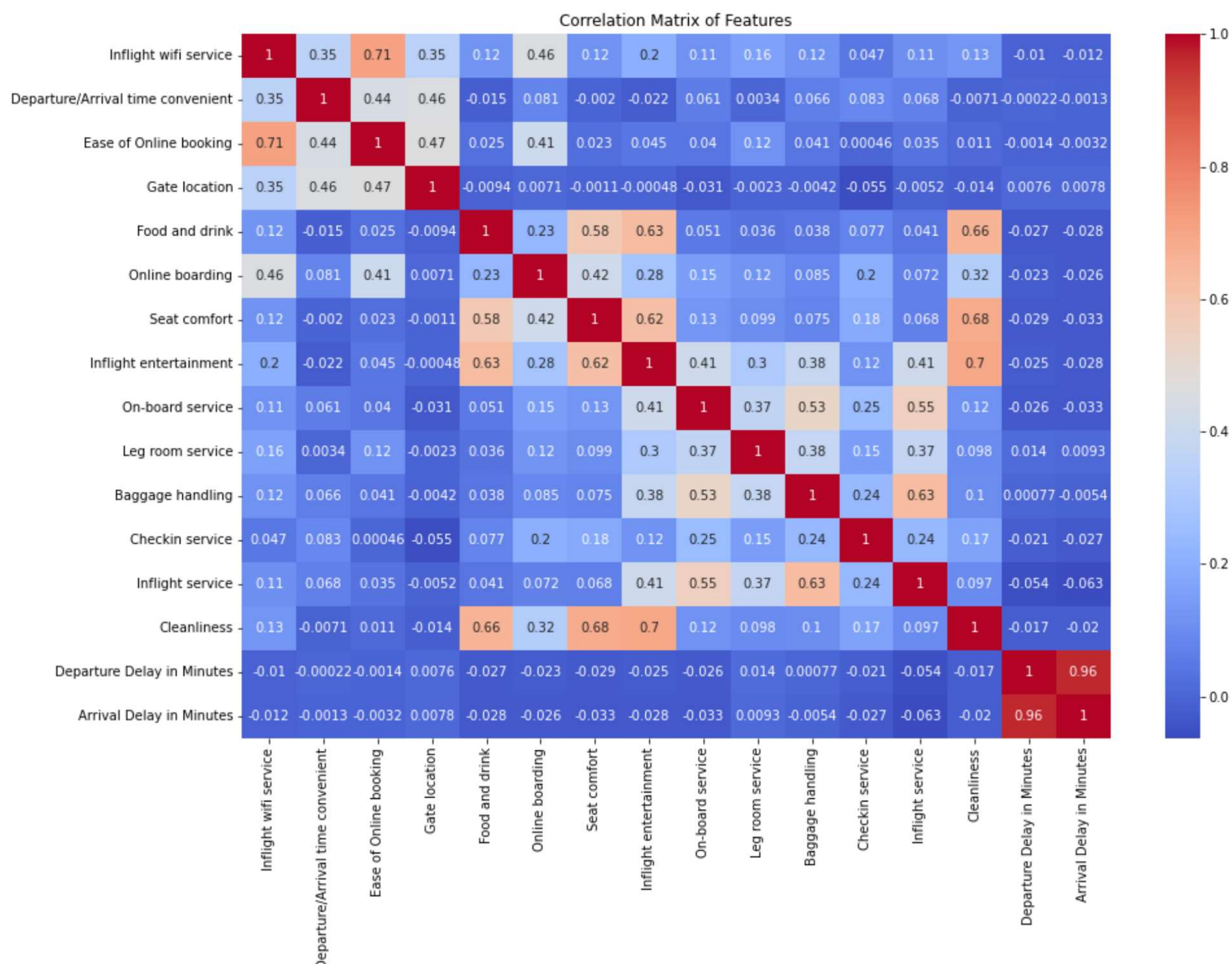
	Ease of Online booking	Gate location	Food and drink	Online boarding \
count	25893.000000	25893.000000	25893.000000	25893.000000
mean	2.755996	2.976442	3.214923	3.261615
std	1.412552	1.281661	1.331895	1.355505
min	0.000000	1.000000	0.000000	0.000000
25%	2.000000	2.000000	2.000000	2.000000
50%	3.000000	3.000000	3.000000	4.000000
75%	4.000000	4.000000	4.000000	4.000000
max	5.000000	5.000000	5.000000	5.000000

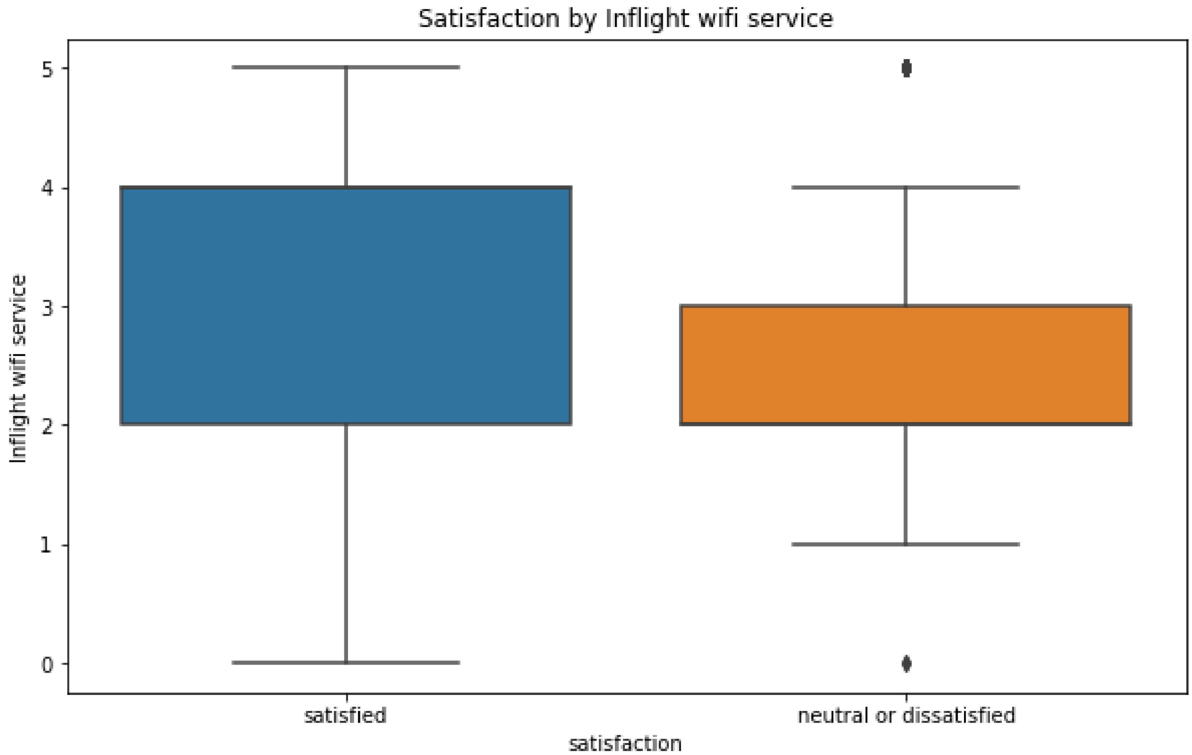
	Seat comfort	Inflight entertainment	On-board service \
count	25893.000000	25893.000000	25893.000000
mean	3.448886	3.356969	3.385587
std	1.320254	1.338643	1.282033
min	1.000000	0.000000	0.000000
25%	2.000000	2.000000	2.000000
50%	4.000000	4.000000	4.000000
75%	5.000000	4.000000	4.000000
max	5.000000	5.000000	5.000000

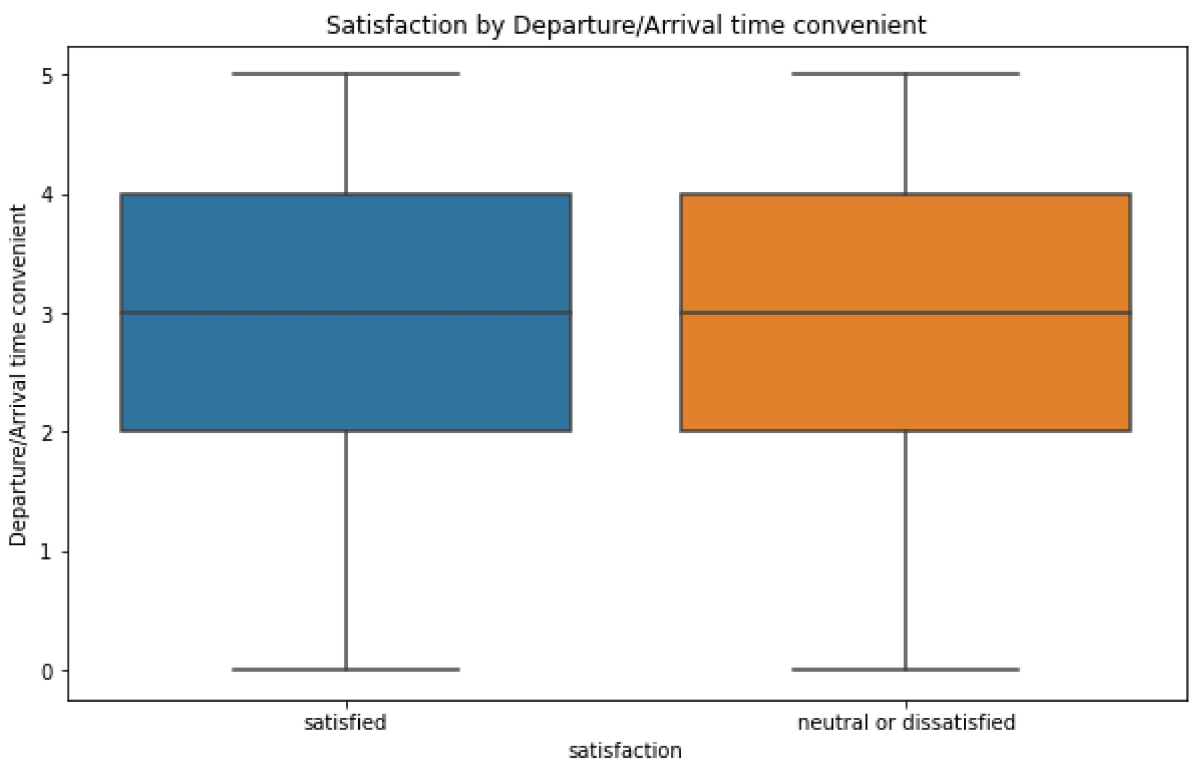
	Leg room service	Baggage handling	Checkin service	Inflight service \
count	25893.000000	25893.000000	25893.000000	25893.000000
mean	3.349786	3.632681	3.313907	3.648824
std	1.319045	1.176220	1.269138	1.180650
min	0.000000	1.000000	1.000000	0.000000
25%	2.000000	3.000000	3.000000	3.000000
50%	4.000000	4.000000	3.000000	4.000000
75%	4.000000	5.000000	4.000000	5.000000
max	5.000000	5.000000	5.000000	5.000000

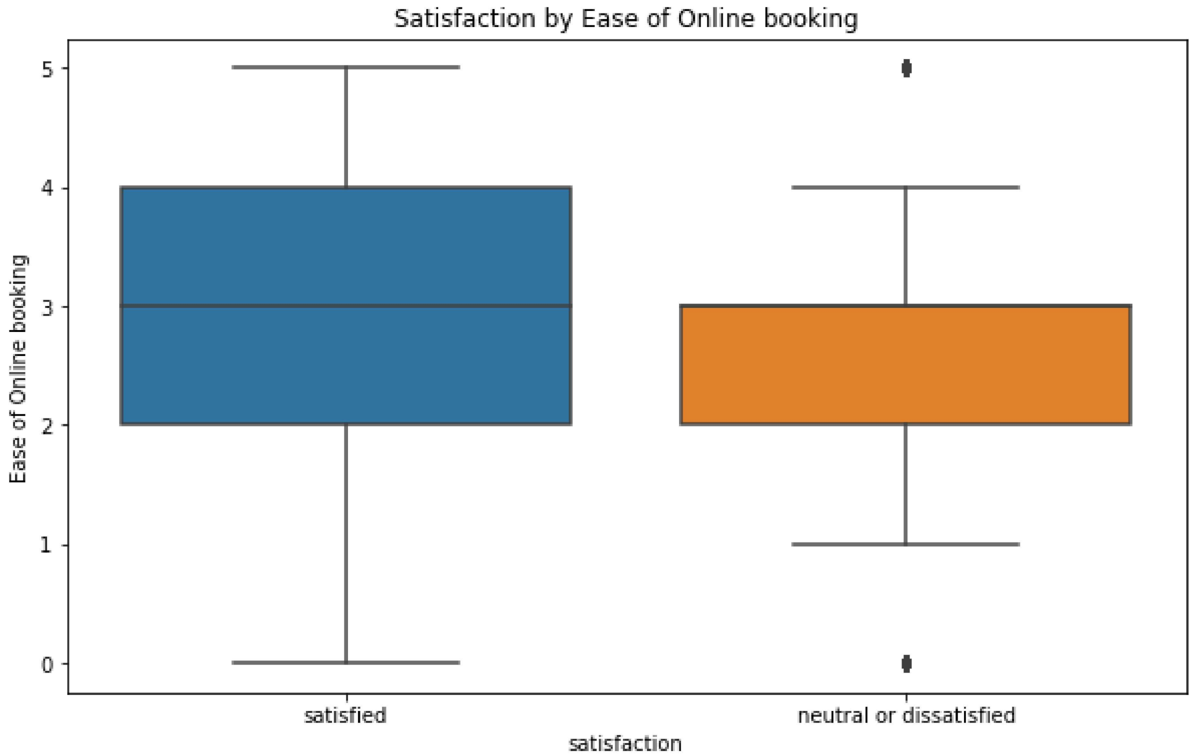
	Cleanliness	Departure Delay in Minutes	Arrival Delay in Minutes
count	25893.000000	25893.000000	25893.000000
mean	3.285521	14.225080	14.740857
std	1.319355	37.185919	37.517539
min	0.000000	0.000000	0.000000
25%	2.000000	0.000000	0.000000
50%	3.000000	0.000000	0.000000
75%	4.000000	12.000000	13.000000
max	5.000000	1128.000000	1115.000000

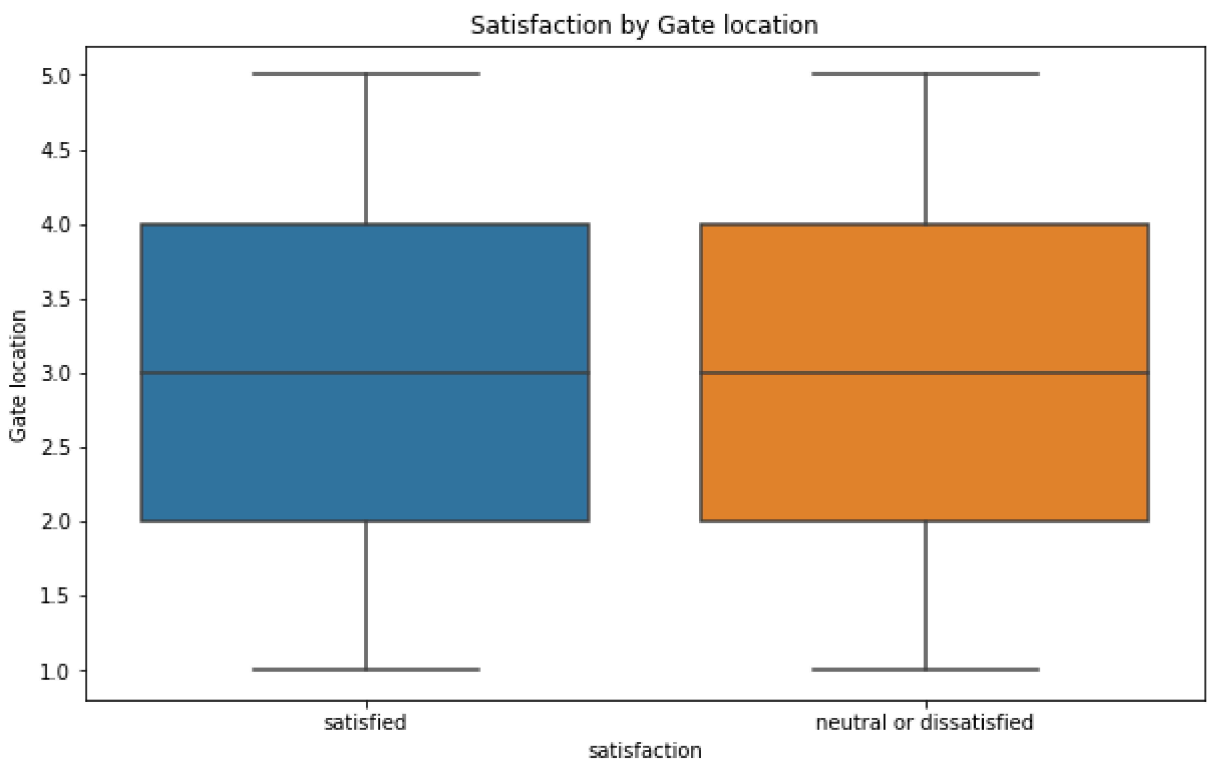

Distribution of Customer Satisfaction
14000
12000
10000
8000
6000
4000
2000
0
count
satisfied
neutral or dissatisfied
satisfaction

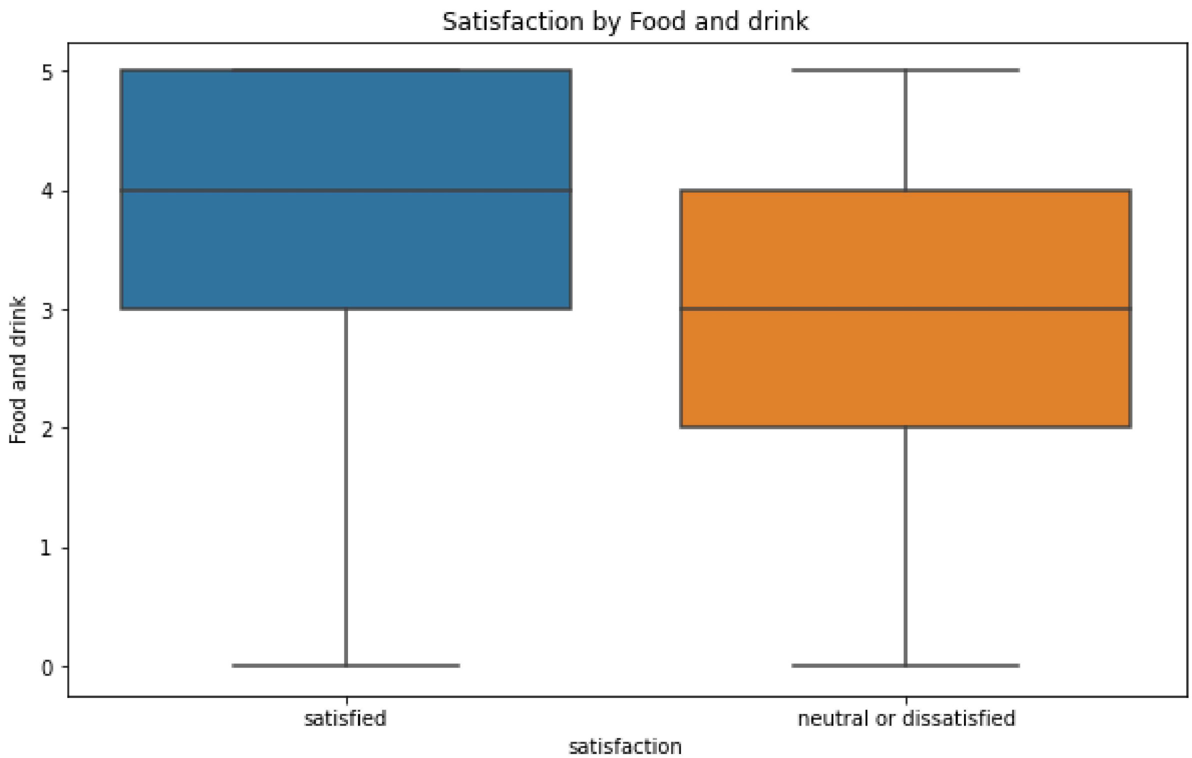

Correlation Matrix of Features
Inflight wifi service
Departure/Arrival time convenient
Ease of Online booking
Gate location
Food and drink
Online boarding
Seat comfort
Inflight entertainment
On-board service
Leg room service
Baggage handling
Checkin service
Inflight service
Cleanliness
Departure Delay in Minutes
Arrival Delay in Minutes
1.0
0.8
0.6
0.4
0.2
0.0

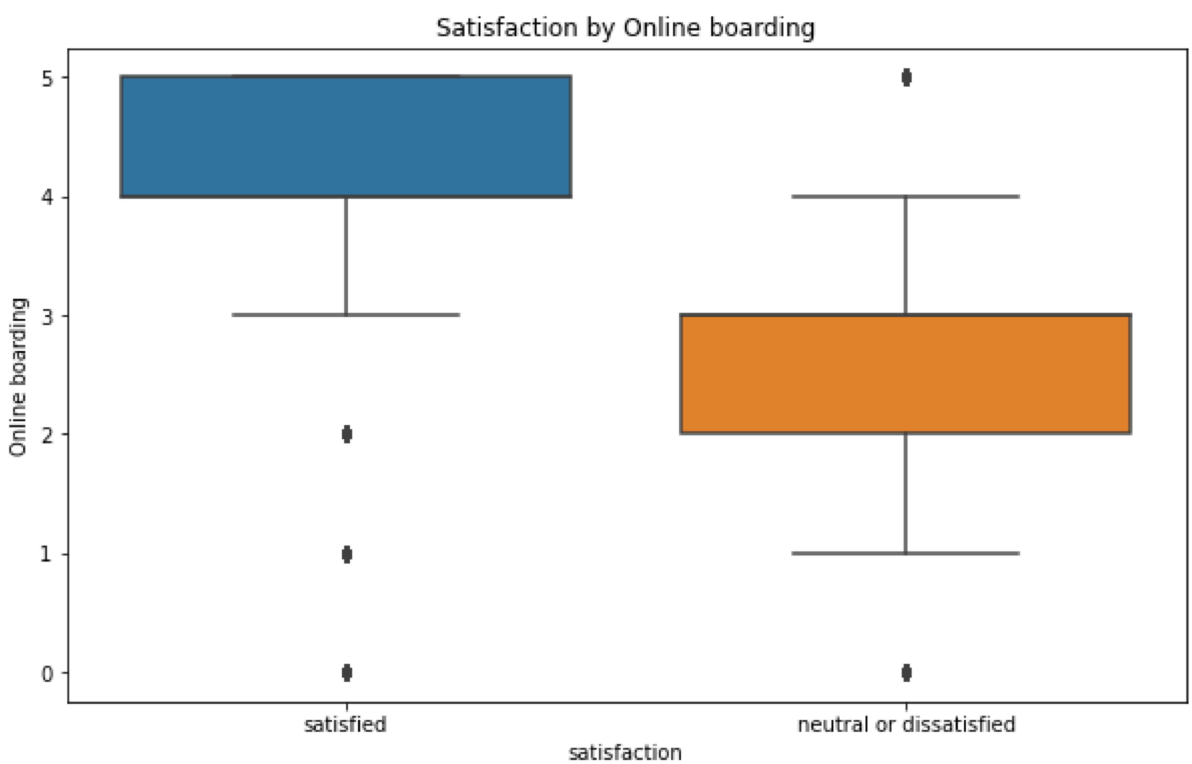

Satisfaction by Inflight wifi service
Inflight wifi service
0
1
2
3
4
5
satisfied
neutral or dissatisfied
satisfaction

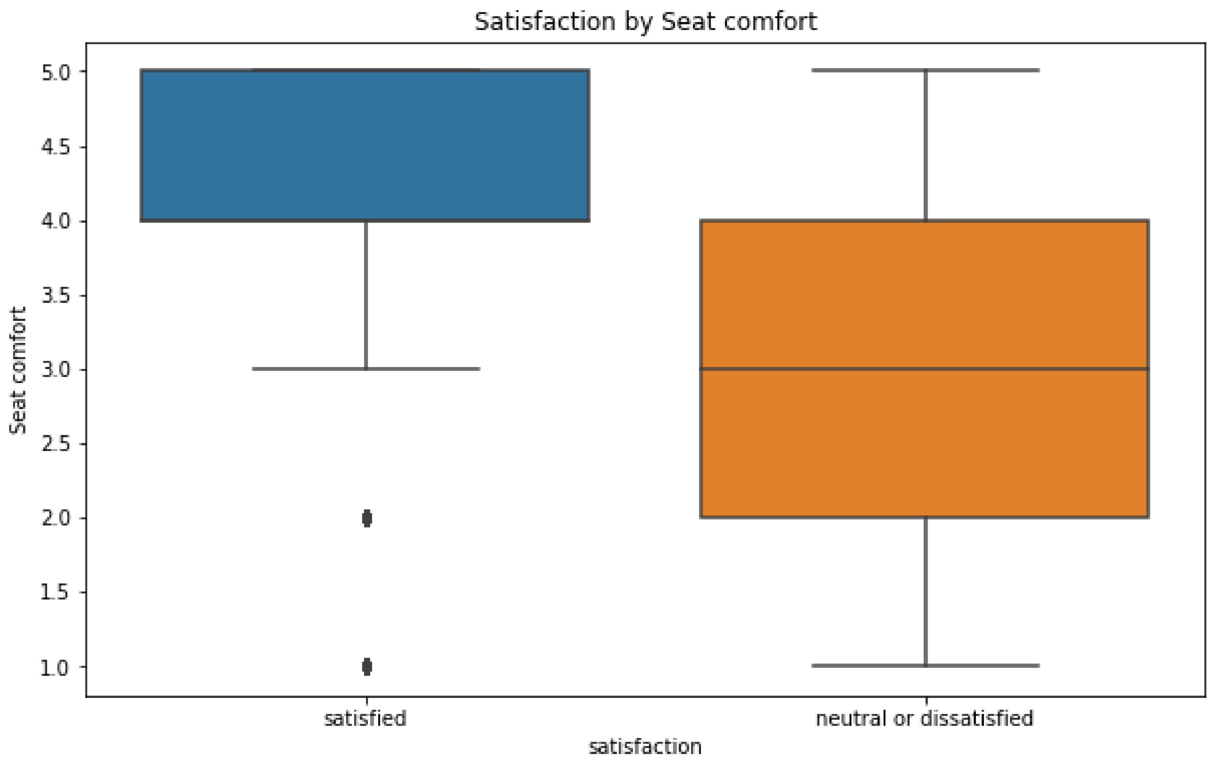

Satisfaction by Departure/Arrival time convenient
Departure/Arrival time convenient
0
1
2
3
4
5
satisfied
neutral or dissatisfied
satisfaction

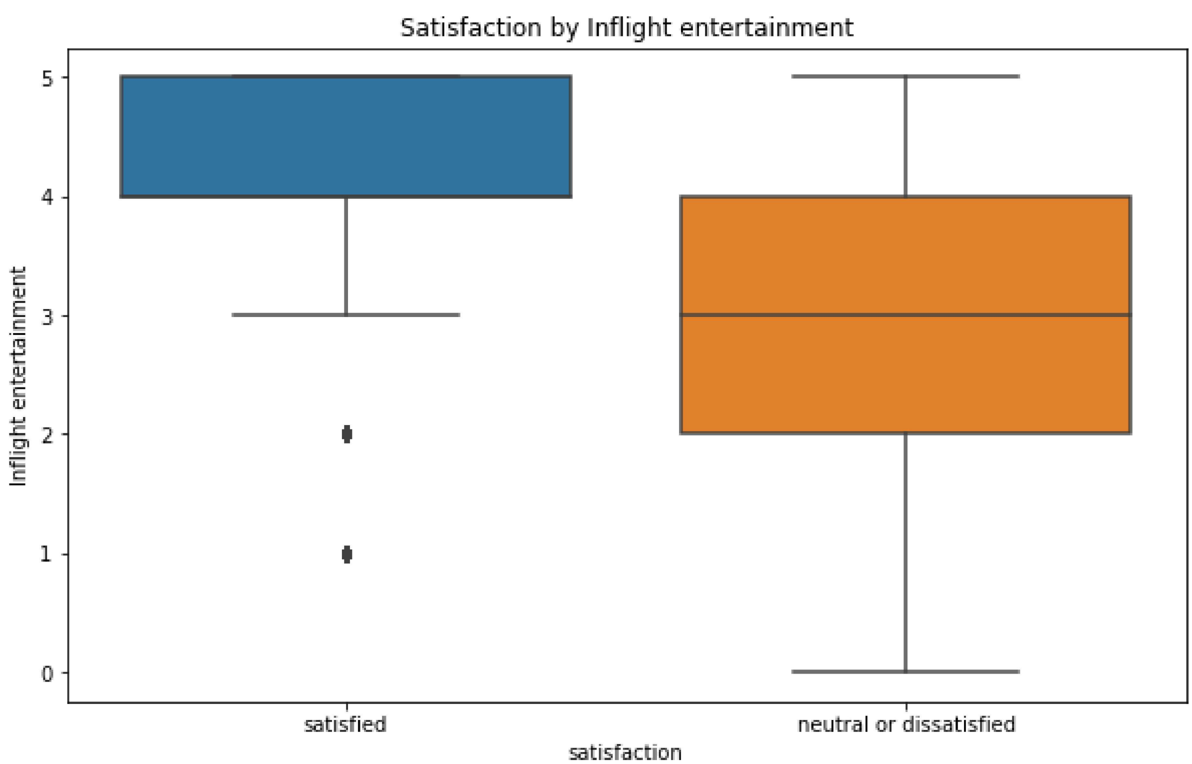

Satisfaction by Ease of Online booking
Ease of Online booking
5
4
3
2
1
0
satisfied
neutral or dissatisfied
satisfaction

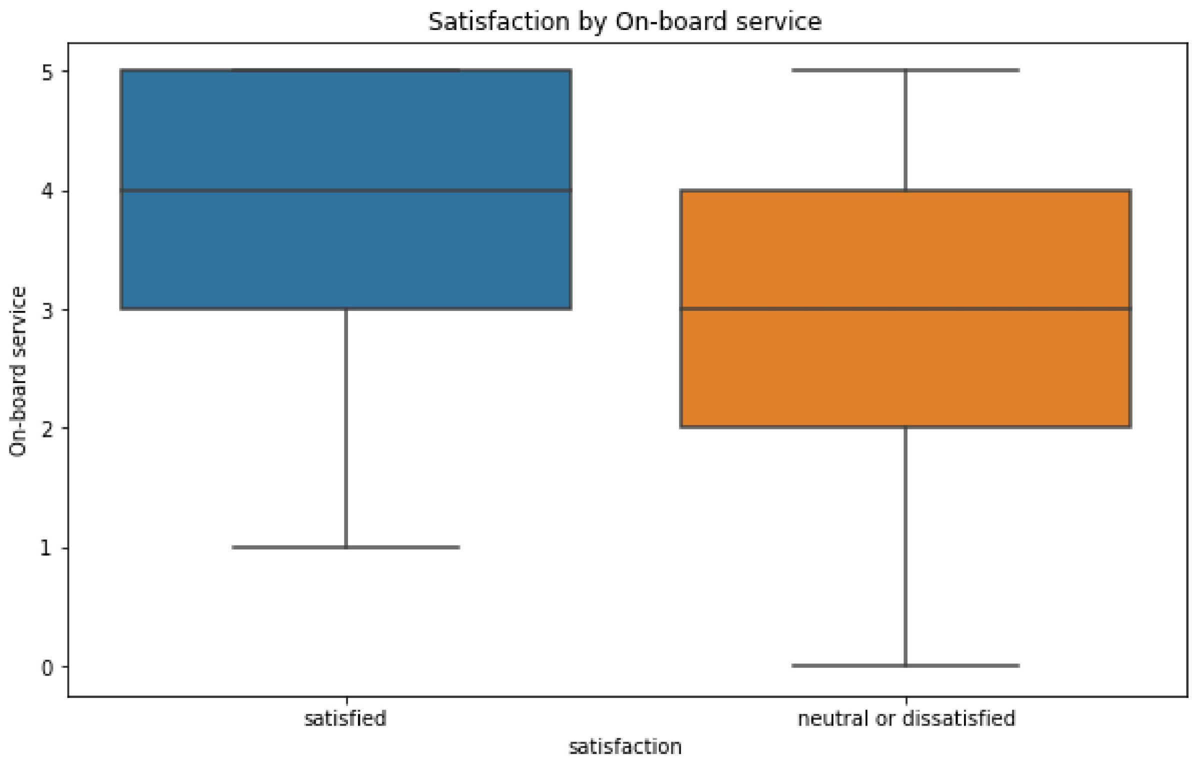

Satisfaction by Gate location
Gate location
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
satisfied
neutral or dissatisfied
satisfaction

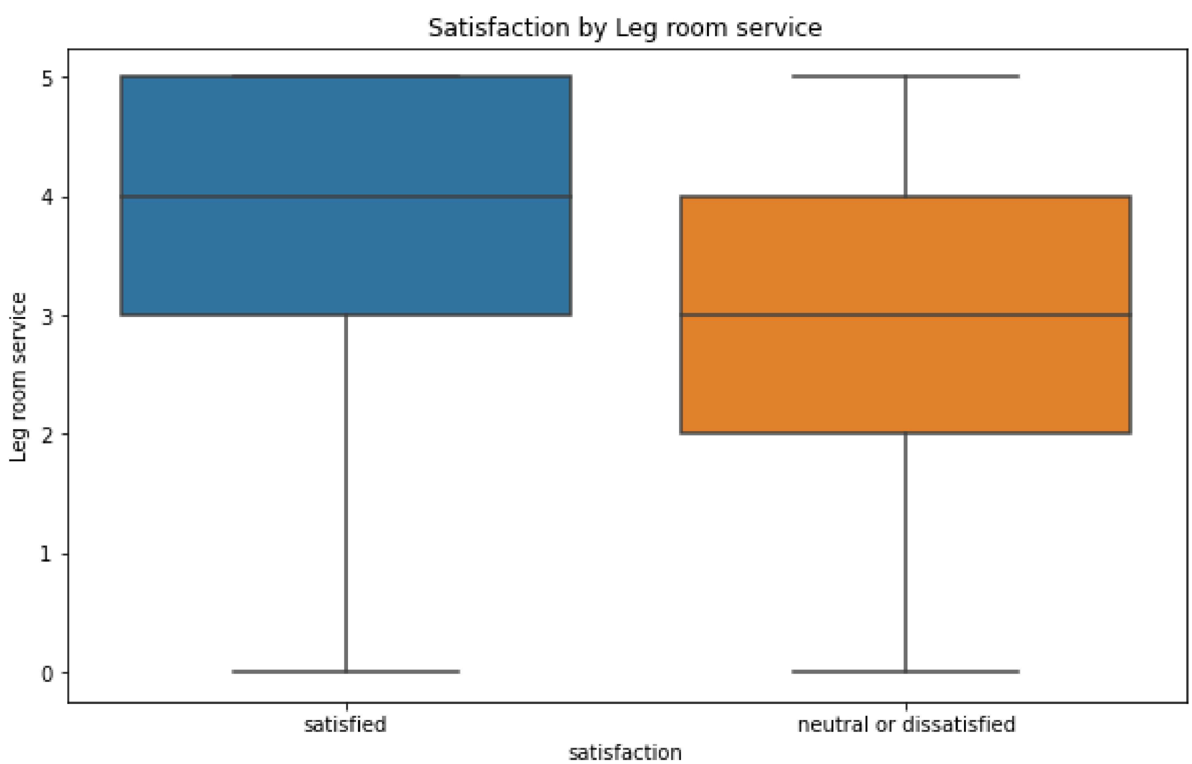

Satisfaction by Food and drink
Food and drink
0
1
2
3
4
5
satisfied
neutral or dissatisfied
satisfaction

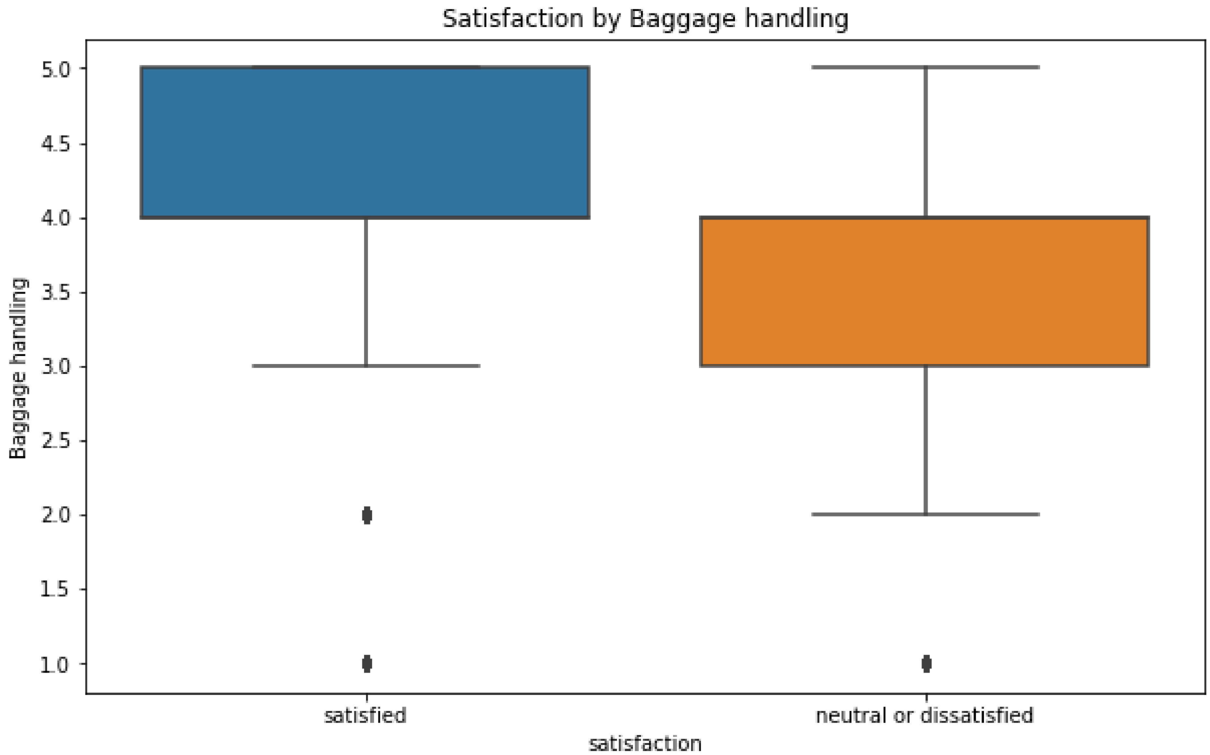

Satisfaction by Online boarding
Online boarding
0
1
2
3
4
5
satisfied
neutral or dissatisfied
satisfaction

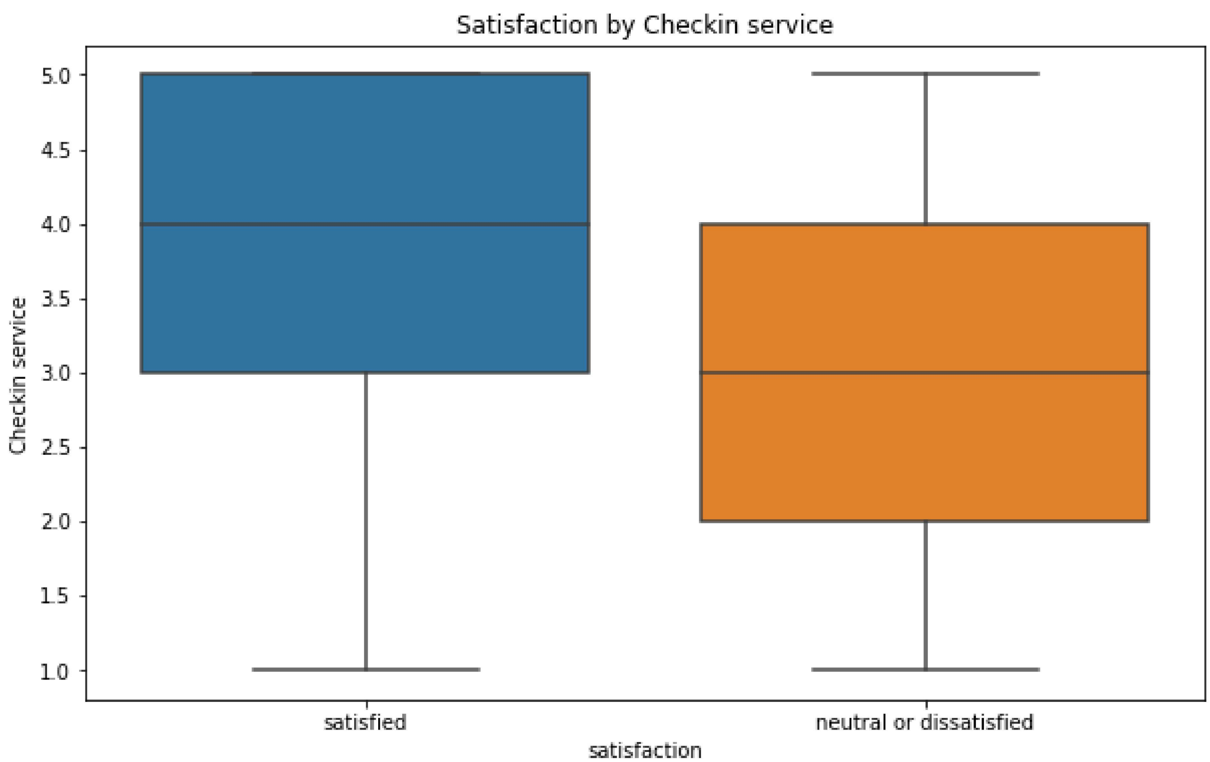

Satisfaction by Seat comfort
Seat comfort
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
satisfied
neutral or dissatisfied
satisfaction

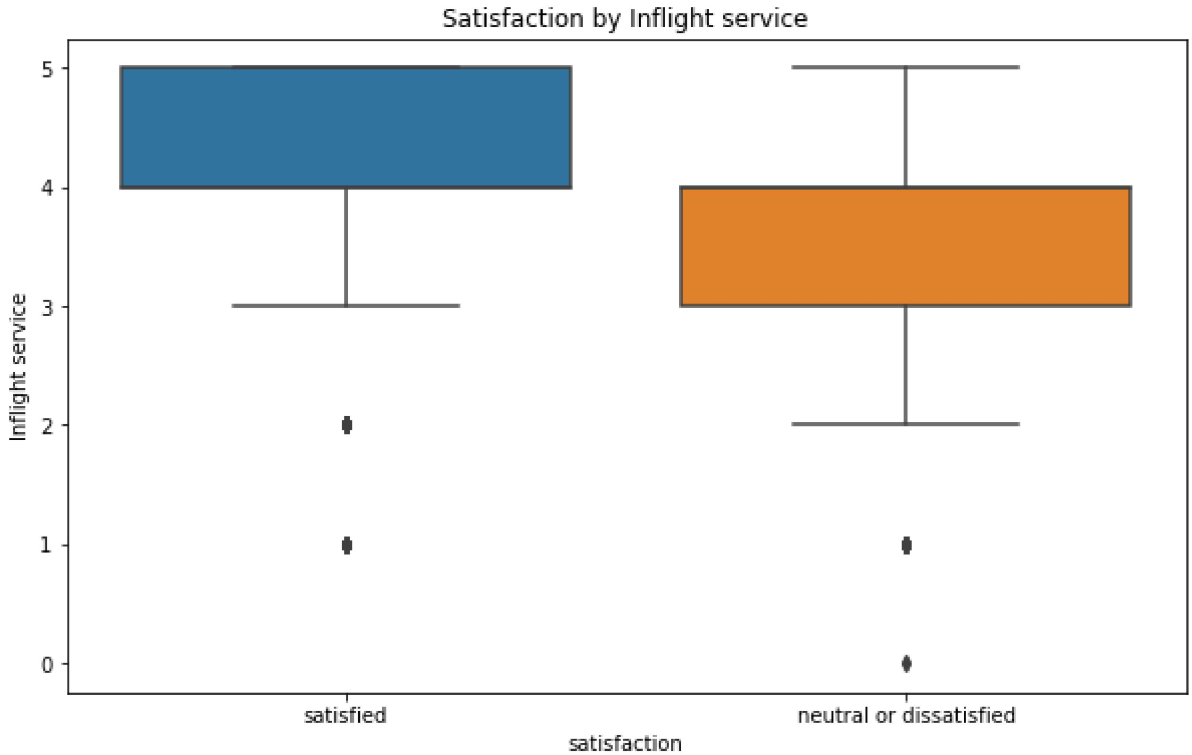

Satisfaction by Inflight entertainment
Inflight entertainment
5
4
3
2
1
0
satisfied
neutral or dissatisfied
satisfaction

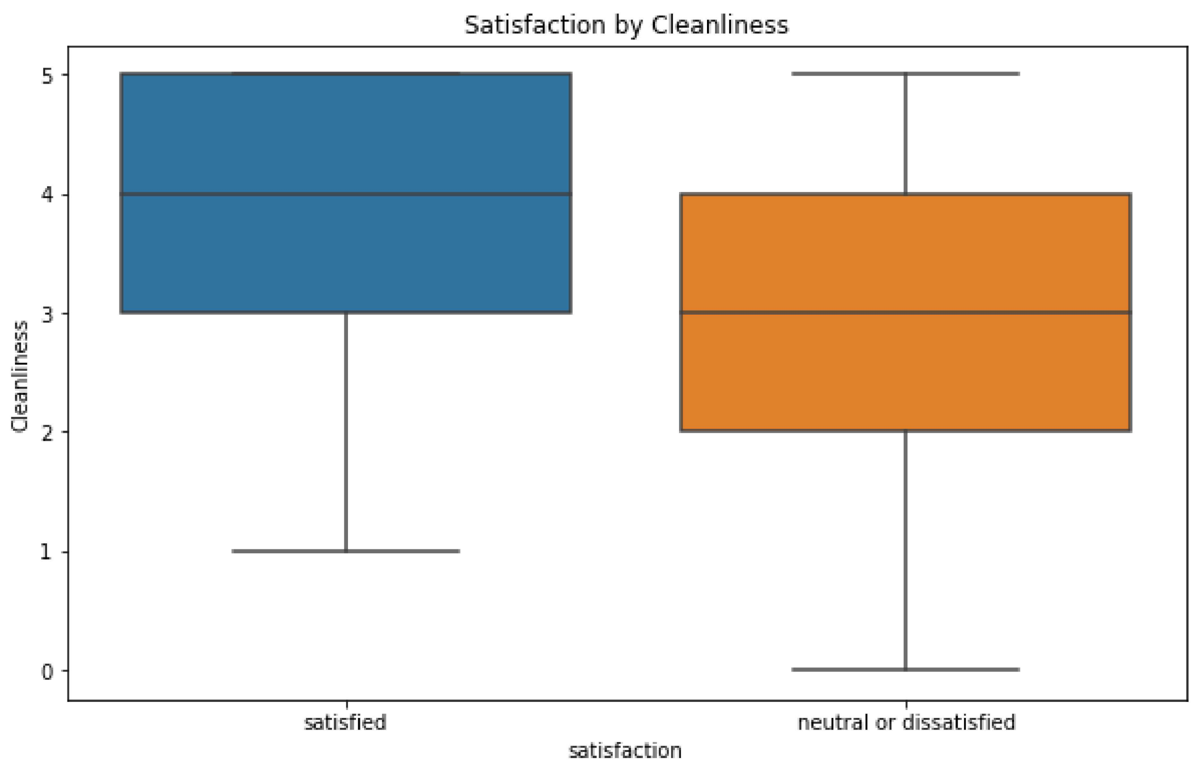

Satisfaction by On-board service
On-board service
0
1
2
3
4
5
satisfied
neutral or dissatisfied
satisfaction

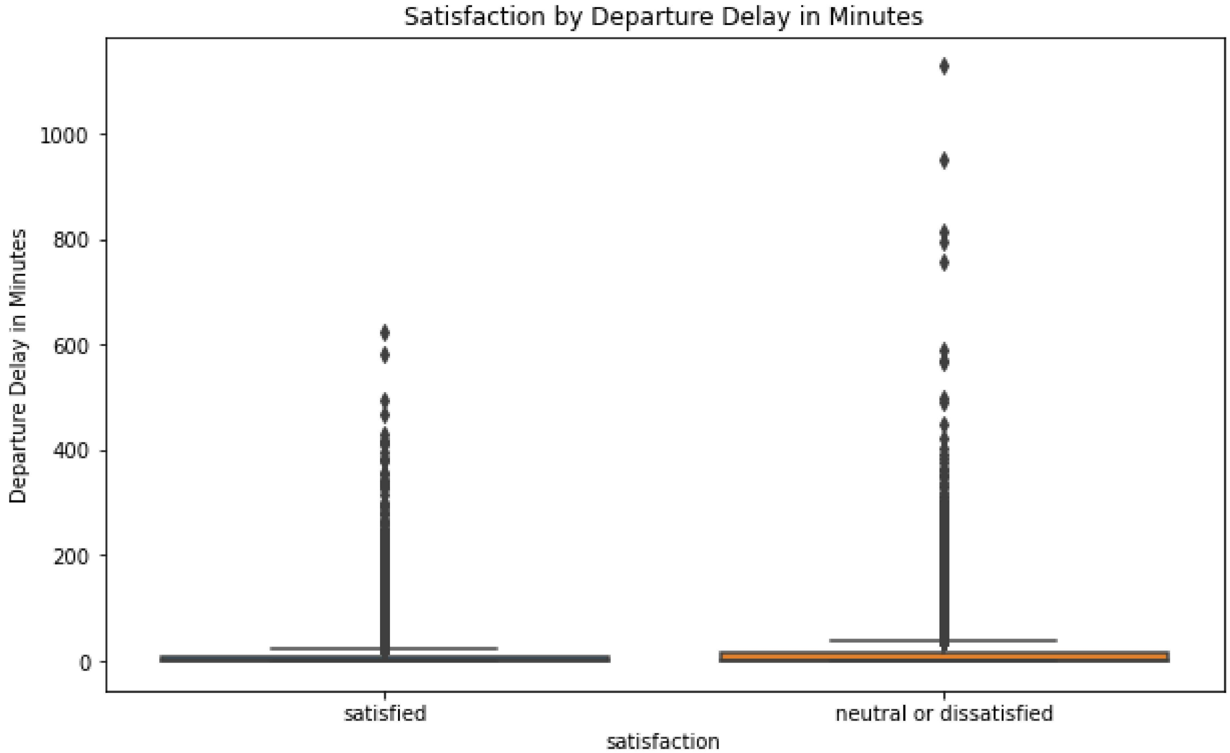

Satisfaction by Leg room service
Leg room service
0
1
2
3
4
5
satisfied
neutral or dissatisfied
satisfaction

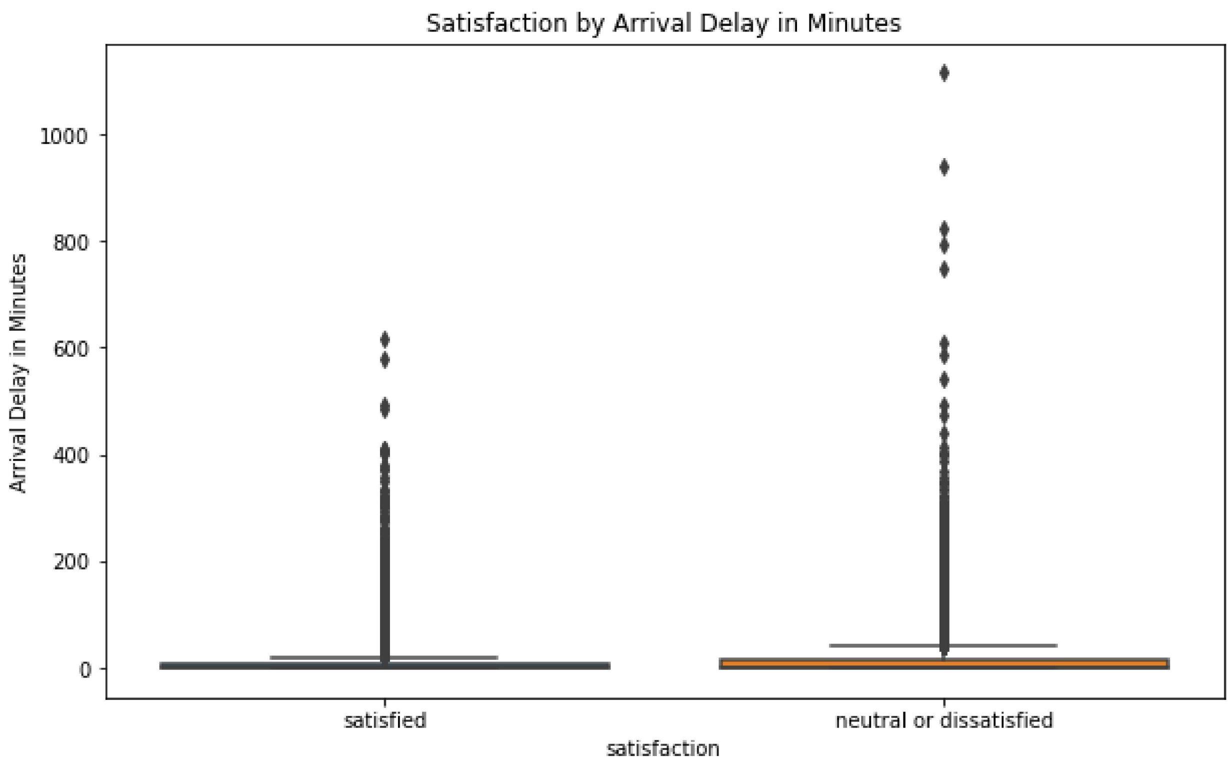

Satisfaction by Baggage handling
Baggage handling
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
satisfied
neutral or dissatisfied
satisfaction


Satisfaction by Checkin service
Checkin service
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
satisfied
neutral or dissatisfied
satisfaction


Satisfaction by Inflight service
Inflight service
0
1
2
3
4
5
satisfied
neutral or dissatisfied
satisfaction


Satisfaction by Cleanliness
Cleanliness
0
1
2
3
4
5
satisfied
neutral or dissatisfied
satisfaction


Satisfaction by Departure Delay in Minutes
Departure Delay in Minutes
0
200
400
600
800
1000
satisfied
neutral or dissatisfied
satisfaction


Satisfaction by Arrival Delay in Minutes
Arrival Delay in Minutes
0
200
400
600
800
1000
satisfied
neutral or dissatisfied
satisfaction

```
In [ ]:
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